

Important Notes

Mobile Manipulator Navigation — MiR250 + UR5e

On the planner experiments

The local-planner comparison study lives in a separate repository: [nav2-local-planner-benchmark](#). It was kept apart from the main project repository because it relies on its own, self-contained simulation environment — a different robot and test world, set up specifically to isolate the planners' behaviour — and bundling it with the mobile-manipulator workspace would have mixed two unrelated environments in one place.

The full results logs for the planner experiments — the per-trial data behind the figures and statistics reported here — are included in that benchmark repository itself, alongside the code used to produce them, so the analysis can be reproduced or re-run on new data.

On the report as a reference

This report is intended to be more than a record of the steps taken. It is written so that someone beginning their own work with ROS 2 and Gazebo could follow it and understand not just *what* was done but *why*. Each part is explained as clearly and accessibly as possible, and the architecture of each system is described before the changes to it.

That ordering is deliberate. Understanding the architecture first — how the description, the controllers, the coordinate tree, the navigation servers, and the costmap layers fit together — is, in my view, the single most important step before starting to build anything. Most of the difficult faults in a project like this are not coding mistakes but consequences of misunderstanding where a piece of behaviour comes from. The report leads with that understanding so that the practical steps which follow make sense and can be adapted rather than merely copied.